A Population objective and empirical estimation bias

A.1 Population-objective identities

The following identities concern fixed candidate parameters and stationary signals with the displayed second moments finite, under the input–innovation independence and model assumptions of Appendix D. Write $\delta\theta = \theta - \theta^\star$.

PL parametrization. Let $\varphi^\star(t)^\top$ denote the ideal PL regressor row formed with the true internal signal $\alpha^\star$, the measured input, and past true disturbances $e^\star$, using the adopted normalization. The system equations give

$$c(t) = \varphi^\star(t)^\top \theta^\star + \nu(t), \qquad \mathbb{E}[\varphi^\star(t)\nu(t)] = 0. \tag{A.1}$$

The orthogonality follows because the signal-path entries are functions of the input, which is independent of the innovation process, while the disturbance-path entries involve past disturbances. Thus, defining $\Gamma_\varphi = \mathbb{E}[\varphi^\star(t)\varphi^\star(t)^\top]$, the ideal population objective satisfies

$$2\mathcal{J}^\star_{\mathrm{PL}}(\theta) := \mathbb{E}[(c(t) - \varphi^\star(t)^\top\theta)^2] = \sigma_\nu^2 + \delta\theta^\top \Gamma_\varphi \delta\theta. \tag{A.2}$$

Indeed, the residual is $\nu(t) - \varphi^\star(t)^\top\delta\theta$, and the cross term vanishes by (A.1). If $\Gamma_\varphi \succ 0$, the unique minimizer is $\theta^\star$. This is the usual innovation-regression argument underlying classical PL identification theory [3].

The implemented PL criterion instead freezes a reconstructed row $\hat\varphi^{(k)}(t)^\top$ of $\hat\Phi^{(k)}$ during an inner update. To state its population identity without assuming perfect reconstruction, fix a row index t and define the following moments whenever they exist (their dependence on t is suppressed):

$$\begin{aligned} \varepsilon^{(k)}(t) &= \nu(t) + [\varphi^\star(t) - \hat\varphi^{(k)}(t)]^\top\theta^\star, \\ R_k &= \mathbb{E}[\hat\varphi^{(k)}(t)\hat\varphi^{(k)}(t)^\top], \qquad q_k = \mathbb{E}[\hat\varphi^{(k)}(t)\varepsilon^{(k)}(t)]. \end{aligned} \tag{A.3}$$

For $\mathcal{J}^{(k)}_{\mathrm{PL}}(\theta) = \frac{1}{2}\mathbb{E}[(c(t) - \hat\varphi^{(k)}(t)^\top\theta)^2]$, expanding the square yields the exact quadratic identity

$$2\mathcal{J}^{(k)}_{\mathrm{PL}}(\theta) = \mathbb{E}[(\varepsilon^{(k)}(t))^2] - 2\delta\theta^\top q_k + \delta\theta^\top R_k \delta\theta. \tag{A.4}$$

The expected record-averaged criterion has the same form, with the moments averaged over the retained time indices. If $R_k \succ 0$, its unconstrained population minimizer is $\theta^\star + R_k^{-1}q_k$. Consequently, the practical PL surrogate is minimized at the truth when $q_k = 0$; ideal reconstruction is a sufficient condition, but joint estimation alone neither guarantees nor violates this orthogonality. The shift $R_k^{-1}q_k$ describes the minimizer of a frozen population surrogate, not the finite-record bias of the complete iterative algorithm. Its evolution across outer iterations requires a separate convergence analysis.

FSM parametrization. For the direct innovation criterion, define

$$\mathcal{J}_{\mathrm{FSM}}(\theta) = \frac{1}{2}\mathbb{E}[r^2_{\nu,\mathrm{FSM}}(t;\theta)], \tag{A.5}$$

$$\Delta\beta(t;\theta) = \beta_{\mathrm{FSM}}(t;\theta^\star) - \beta_{\mathrm{FSM}}(t;\theta), \tag{A.6}$$

$$\begin{aligned} d &= [d_1, \dots, d_{n_d}]^\top, \qquad E^\star(t) = [e^\star(t-1), \dots, e^\star(t-n_d)]^\top, \\ \Gamma_e &= \mathbb{E}[E^\star(t)E^\star(t)^\top]. \end{aligned}$$

For an exact signal representation, the residual decomposition in Appendix D gives

$$r_{\nu,\mathrm{FSM}}(t;\theta) = \nu(t) + (d - d^\star)^\top E^\star(t) + D(z^{-1};\theta)\Delta\beta(t;\theta).$$

The cross terms vanish by input–innovation independence and orthogonality of the current innovation to past disturbances. Hence

$$2\mathcal{J}_{\mathrm{FSM}}(\theta) = \sigma_\nu^2 + (d - d^\star)^\top \Gamma_e (d - d^\star) + \mathbb{E}\Big[\big(D(z^{-1};\theta)\Delta\beta(t;\theta)\big)^2\Big]. \tag{A.7}$$

With $\Gamma_e \succ 0$ and an identifiable normalized signal parametrization, $\theta^\star$ uniquely minimizes this ideal population objective.

For a finite FSM approximation, $\Delta\beta$ must instead denote the true deterministic signal minus $\beta_{\mathrm{FSM}}(t;\theta)$. The identity retains its form, but this mismatch need not vanish at the physical parameter $\theta^\star$. Thus the practical PL analysis must account for regressor reconstruction, whereas the FSM analysis must account for its finite signal representation. Both also require consistent treatment of finite-record initial states. Neither ideal population identity, by itself, establishes finite-record unbiasedness.

A.2 Scope of an analytical bias expression

PL identification has a substantial analytical literature. Classical work studies convergence points and local and global convergence of recursive least-squares, generalized and extended least-squares, and maximum-likelihood schemes [6, 3]. Ljung [2] also relates prediction-error methods to approximate methods with simplified gradient calculations. These results justify method-specific convergence and asymptotic analysis; they do not imply that every reconstructed-regressor update minimizes the same nonlinear prediction-error objective.

Analytical bias mechanisms are also available for particular regression models. Söderström [5] treats regressor contamination in errors-in-variables identification. Janczak and Korbicz [1] analyse regressor–disturbance correlation in a transformed polynomial Wiener model, with a disturbance placement different from the present output–AR model. Ni et al. [4] combine reconstruction of unmeasured signals with instrumental-variable conditions for consistent Hammerstein–Wiener identification in coloured output noise. These references support analysing the relevant moment conditions; they do not establish bias merely from the use of reconstructed signals.

For a specified iterative method $\mathcal{A}$, write its returned estimate as

$$\widehat{\theta}_N^{\mathcal{A}} = \mathcal{A}_N(\mathcal{Z}_N;\xi), \qquad b_N^{\mathcal{A}}(\xi) = \mathbb{E}[\widehat{\theta}_N^{\mathcal{A}}] - \theta^\star, \tag{A.8}$$

where $\mathcal{Z}_N = \{u(t), c(t)\}_{t=1}^N$ and ξ collects the initialization, update order, step sizes, regularization, loss target, stopping rules, iteration budget, and FSM resolution. The expectation is well defined whenever the implementation returns an integrable estimate.

The difficulty here is deriving a full finite-record expression for this complete, tunable data-to-estimate map. Its output depends nonlinearly on all preceding reconstructions and updates, as well as any active constraints and data-dependent stopping decisions. The quadratic PL identities above do not remove this dependence because the regressor itself changes across iterations. Likewise, the FSM population identity does not determine the bias of a finite-iteration nonlinear fit. Reaching a prescribed loss target need not mean that either population minimum has been attained.

Consequently, no common closed-form finite-record bias formula follows for all the implemented PL, FSM, gradient-root, and loss-root methods. This is not a claim that analytical bias analysis is impossible or that PL theory is unavailable. Special cases and suitably regular converged estimators can admit exact expressions or asymptotic expansions under additional assumptions. The study below instead quantifies the mean errors of the actual implementations under their configured tuning and stopping rules.

A.3 Record-length study

The two benchmark systems were evaluated using $R = 100$ independent realizations and

$$N \in \{500, 1000, 2000, 5000, 10000\}.$$

Within each implementation, shorter records were nested prefixes of the same realization and were shared by the methods. The PL and FSM implementations used separate random-number generators; their records are not assumed to be paired across parametrizations. All fits started from $\theta^{(0)} = (0, 0, 1, 0, 0, 0)^\top$ and used a maximum of 300 iterations, with the configured method-specific stopping rules. The FSM settings were $(\tau_g, \tau_h) = (128, 128)$ with 1024 frequency samples and $(256, 256)$ with 2048 frequency samples. Data were generated by recursive filtering, with 100 burn-in samples, independently of these FSM approximation settings; fitting used the zero-state convention.

Only the 128-coefficient FSM figures are displayed. The omitted 256-coefficient CSTR curves are visually indistinguishable from those shown for 128 coefficients; the well-behaved Example 1 curves are also virtually equal, particularly at the larger record lengths. Some short-record outlier magnitudes in Example 1 differ substantially, so this visual agreement is not asserted for every method and record length. Both resolutions remain in Table 1.

For each parameter, method, resolution, and record length, let $\mathcal{I}_N$ index the finite returned estimates and $R_N = |\mathcal{I}_N|$. The reported statistics are

$$\widehat{b}_{j,N} = \frac{1}{R_N} \sum_{\ell \in \mathcal{I}_N} \left(\widehat{\theta}^{(\ell)}_{N,j} - \theta^\star_j \right), \tag{A.9}$$

$$\widehat{\mathrm{RMSE}}_{j,N} = \left[\frac{1}{R_N} \sum_{\ell \in \mathcal{I}_N} \left(\widehat{\theta}^{(\ell)}_{N,j} - \theta^\star_j \right)^2 \right]^{1/2}.$$

The error bars are pointwise approximate 95% Monte Carlo intervals $\widehat{b}_{j,N} \pm 1.96\, s_{j,N}/\sqrt{R_N}$, where $s_{j,N}$ is the sample standard deviation of the retained estimates. They quantify uncertainty in the estimated mean error, not uncertainty in a parameter estimate from one record, and are not simultaneous bands. Record length is displayed on a logarithmic axis in all panels; RMSE also uses a logarithmic vertical axis, whereas signed mean errors remain on a linear vertical axis. The mean-error vertical axes are capped at $[-1, 1]$, with tighter automatic ranges retained when possible. Points and error-bar portions outside the displayed range are clipped for visualization only; the underlying statistics are unchanged.

Finite outputs are retained even when the method stalls or reaches the iteration limit. When failures are excluded, the corresponding summaries describe mean error conditional on a finite returned output. Table 1 reports the largest-record outcome counts; *converged* means that the implemented stopping criterion was satisfied, not that a global minimum or unbiased estimate was certified. RGLS for FSM and GHAM-2-H for PL were skipped.

For Example 1, the PL signal-parameter mean errors are small at the largest tested record length (Fig. 1). At $N = 10000$, all five signal-parameter intervals include zero for every available PL method. The d_1 intervals also include zero for RGLS, WS-GNI, the two diagonal-Hessian GHAM methods, and LHAM-3-TIK. In contrast, the d_1 mean errors for WS-GGI and GHAM-2-I remain approximately 0.272 and 0.241, respectively; all their fits reach the iteration limit. These offsets therefore describe the finite-budget implementations and cannot be attributed solely to statistical bias of a converged estimator.

The Example 1 FSM results (Fig. 2) show decreasing RMSE for the loss-root methods. For example, with 128 coefficients, LHAM-3-TIK reduces the d_1 RMSE from 0.119 at $N = 500$ to 0.0187 at $N = 10000$; the final mean error is -0.00218, with interval $[-0.00583, 0.00147]$. Both loss-root methods satisfy their stopping criteria in all 100 largest-record fits, and their six parameter intervals include zero. This is compatible with small mean errors at the achieved Monte Carlo precision, but does not prove unbiasedness.

Other FSM outcomes require caution. Short records produce extreme finite outliers for WS-GNI and GHAM-2-H, particularly at the lower resolution. Values outside the displayed mean-error range are clipped; the logarithmic RMSE axes retain their full magnitude together

with the smaller errors of the other methods. For instance, the $N = 500$, 128-coefficient WS-GNI estimate of b_1 has an RMSE of about 1.38×10^{14} despite all 100 outputs being finite. Furthermore, only 7 of the 100 Example 1 GHAM-2-I fits remain finite at $N = 10000$, at either resolution. Its displayed mean errors and RMSEs are thus conditional on those seven fits and must be interpreted together with the 93 failures.

For the CSTR benchmark, the PL results (Fig. 3) again show small signal-parameter mean errors at the largest record, while the disturbance coefficient is more difficult to estimate. For RGLS, the d_1 mean error decreases from 0.125 to 0.0119 as N increases from 500 to 10000, and the RMSE decreases from 0.172 to 0.0206. Nevertheless, its final interval $[0.00860, 0.0152]$ excludes zero, although all 100 fits are marked converged. WS-GNI and the diagonal-Hessian GHAM methods give essentially the same largest-record statistics. WS-GGI and GHAM-2-I retain larger d_1 offsets, approximately 0.218 and 0.370, respectively, with every fit reaching the iteration limit.

The CSTR FSM results (Fig. 4) exhibit detectable signal-parameter offsets despite decreasing variability. At $N = 10000$, LHAM-3-TIK has mean errors approximately -0.00520 for a_1, 0.00636 for b_2, and 0.0174 for d_1. All six parameter intervals exclude zero for both loss-root methods, although every fit satisfies its stopping criterion at every tested record length. The two FSM resolutions give essentially identical CSTR statistics, and the largest-record Example 1 loss-root results also agree closely. Thus increasing the tested resolution does not remove the CSTR offsets; this comparison alone neither identifies their cause nor rules out every approximation effect.

Overall, increasing the record length reduces many errors, but does not eliminate all observed method-dependent offsets under the configured computational budget. The results establish finite-record behavior of the returned estimates. They do not establish exact unbiasedness when an interval contains zero, or inconsistency when an interval excludes zero at the tested record lengths.

Table 1: Outcomes at $N = 10000$. Entries give finite outputs / outputs satisfying the configured stopping criterion, out of 100 trials. A dash denotes a skipped method. All finite outputs enter the mean-error and RMSE summaries.

Method	PL	FSM, 128	FSM, 256
Example 1			
RGLS	100/99	—	—
WS-GNI	100/0	100/0	100/0
WS-GGI	100/0	99/0	100/0
GHAM-1-dH	100/0	100/0	100/0
GHAM-2-dH	100/0	100/0	100/0
GHAM-2-H	—	100/0	100/0
GHAM-2-I	100/0	7/0	7/0
LHAM-1-TIK	100/0	100/100	100/100
LHAM-3-TIK	100/53	100/100	100/100
CSTR			
RGLS	100/100	—	—
WS-GNI	100/40	100/0	100/0
WS-GGI	100/0	100/0	100/0
GHAM-1-dH	100/0	100/0	100/0
GHAM-2-dH	100/0	100/0	100/0
GHAM-2-H	—	100/0	100/0
GHAM-2-I	100/0	100/0	100/0
LHAM-1-TIK	100/0	100/100	100/100
LHAM-3-TIK	100/59	100/100	100/100

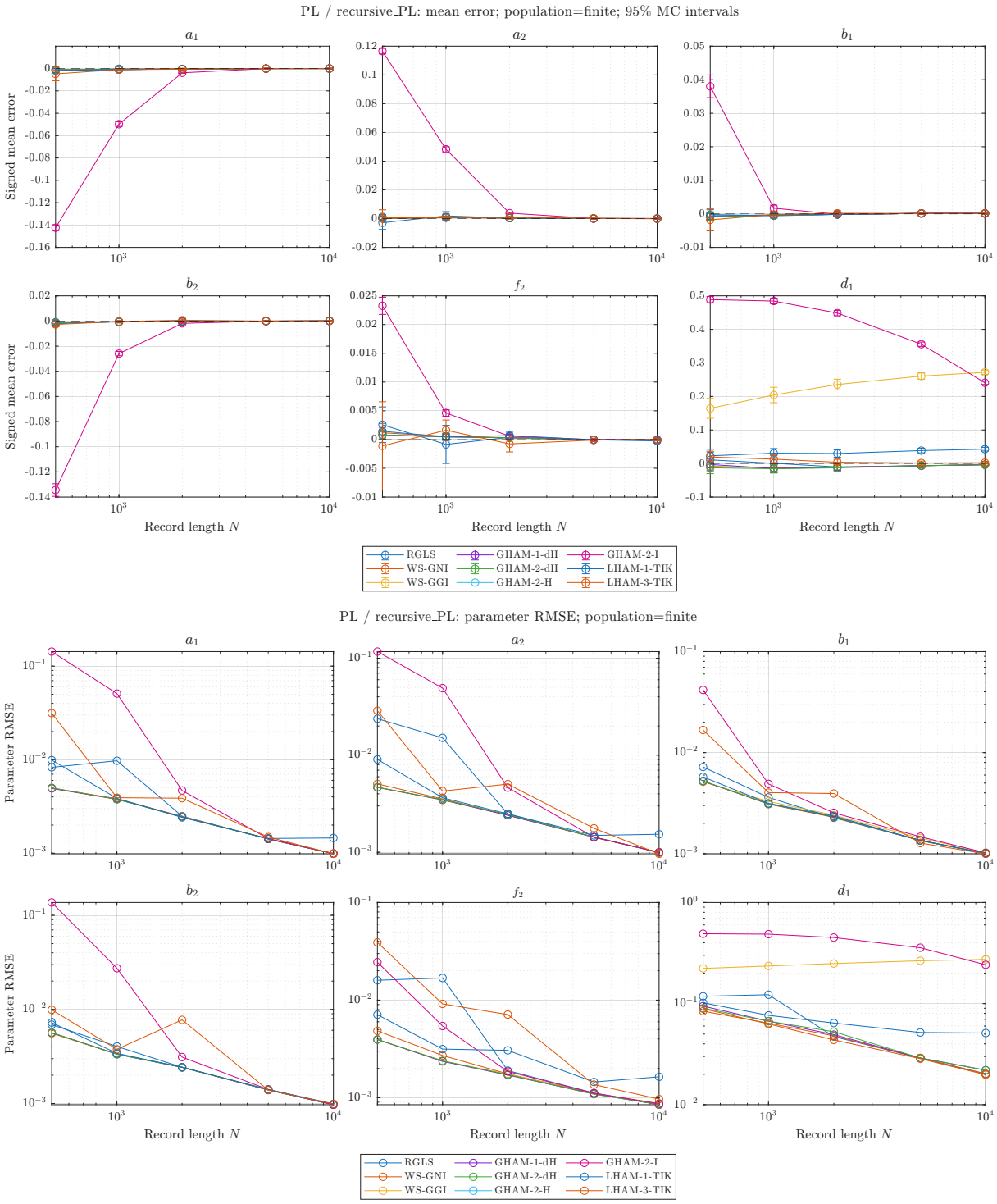


Figure 1: Example 1, PL: signed mean parameter errors with pointwise 95% Monte Carlo intervals (upper panel, linear vertical scale), and parameter RMSEs (lower panel, logarithmic vertical scale). Mean-error axes are capped at $[-1, 1]$. Finite stalled and iteration-limited outputs are retained.

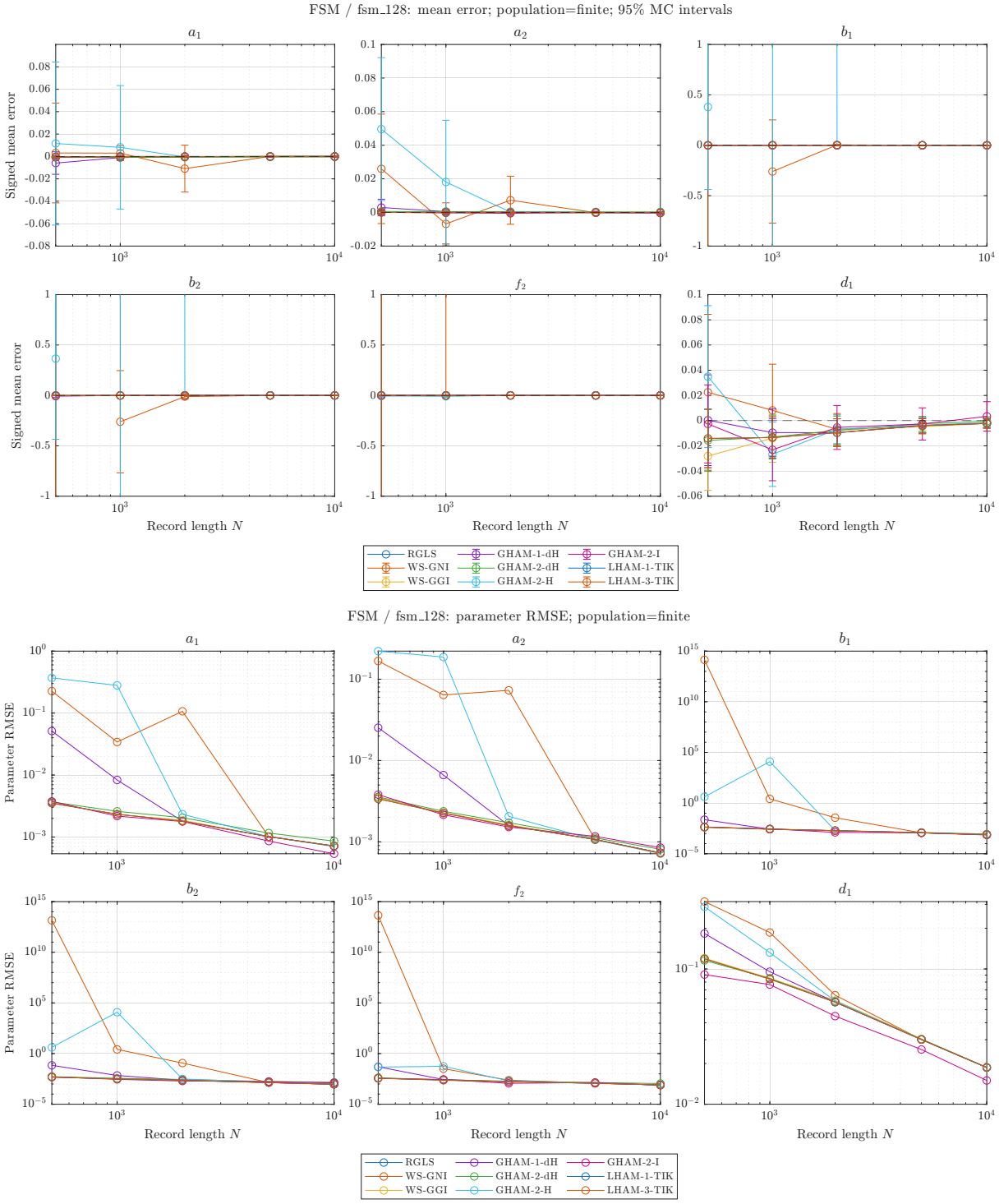


Figure 2: Example 1, FSM with 128 coefficients and 1024 frequency samples: signed mean errors and pointwise 95% Monte Carlo intervals (upper panel, linear vertical scale), and RMSEs (lower panel, logarithmic vertical scale). Mean-error axes are capped at $[-1, 1]$; outlying values and error-bar portions are clipped. Summaries exclude failed fits.

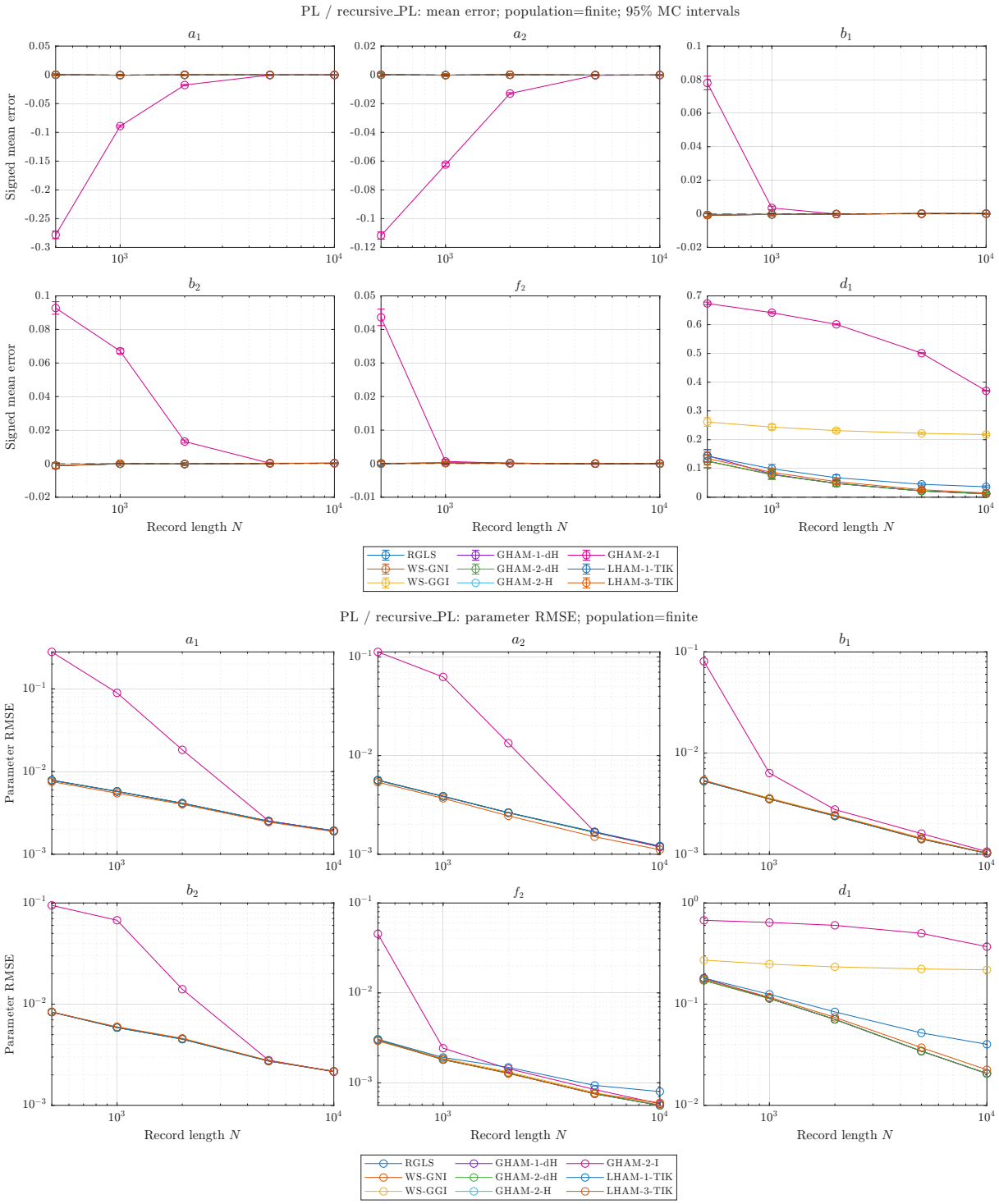


Figure 3: CSTR, PL: signed mean parameter errors with pointwise 95% Monte Carlo intervals (upper panel, linear vertical scale), and parameter RMSEs (lower panel, logarithmic vertical scale). Mean-error axes are capped at $[-1, 1]$. Finite stalled and iteration-limited outputs are retained.

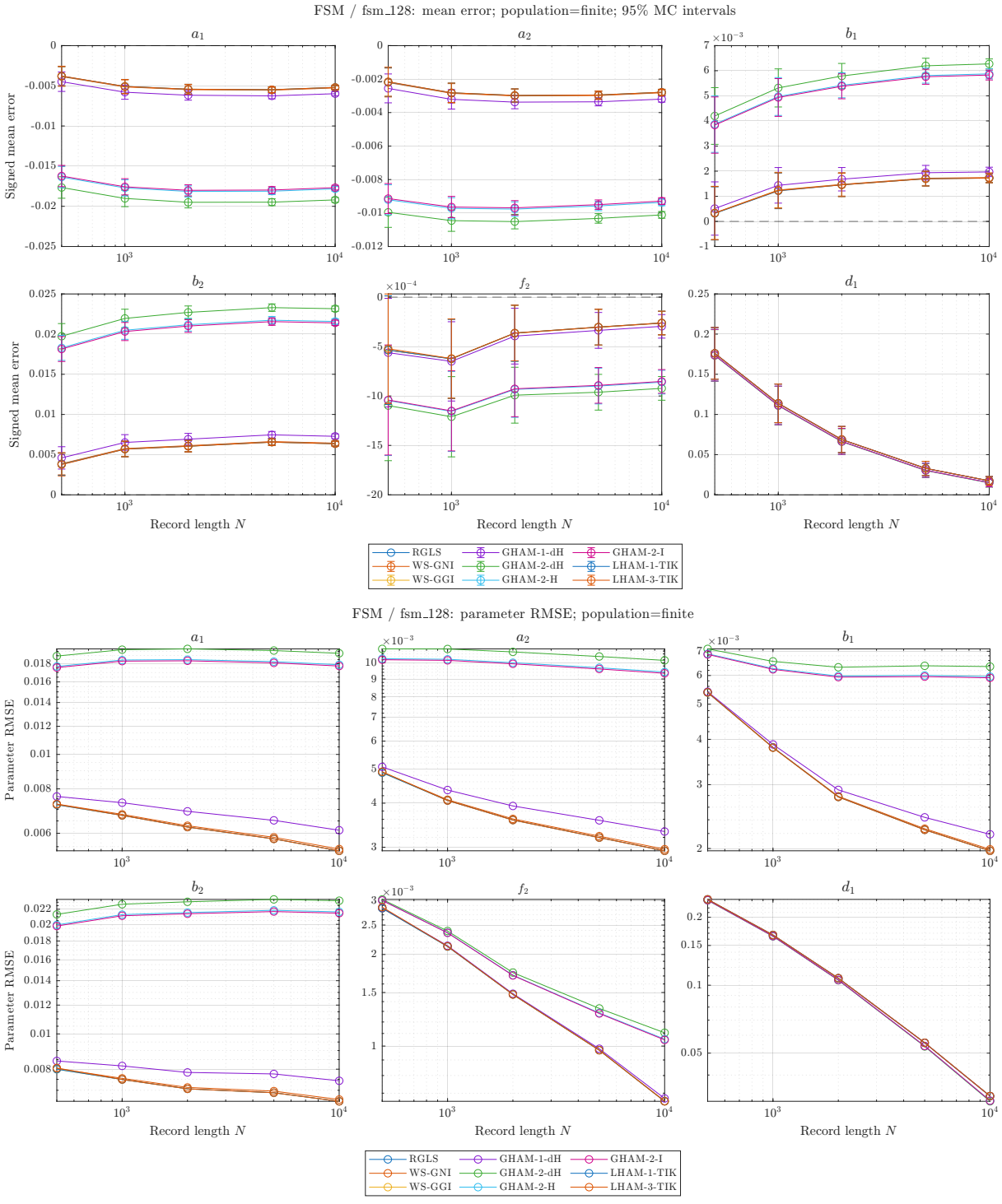


Figure 4: CSTR, FSM with 128 coefficients and 1024 frequency samples: signed mean errors and pointwise 95% Monte Carlo intervals (upper panel, linear vertical scale), and RMSEs (lower panel, logarithmic vertical scale). Mean-error axes are capped at $[-1, 1]$. Finite iteration-limited outputs are retained.

References

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